HW 4

Darin Peries

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Link to Git repo

Problem 1

Prove that $\|v\|_B := \sqrt{v^*Bv}$ defines a norm on $\mathbb{C}^n$ given that B is both positive definite and hermitian.

Proof for homogeneity

Given $\lambda \in \mathbb{C}$ and $x \in \mathbb{C}^n$. $\|\lambda x\|_B = \sqrt{(\lambda^* x^*)B(\lambda x)} = \sqrt{|\lambda|^2 x^*Bx} = |\lambda|\sqrt{x^*Bx} = |\lambda|\|x\|_B$. Therefore the norm maintains homogeneity.

Proof for triangle inequality

Given $x,\ y \in \mathbb{C}^n$, $\|x+y\|_B = \sqrt{(x+y)^*B(x+y)} = \sqrt{(x^*+y^*)B(x+y)} = \sqrt{(x^*B+y^*B)(x+y)} = \sqrt{x^*Bx + x^*By + y^*Bx + y^*By}$. Because matrix B is hermitian, $(x^*By)^* = y^*Bx$. Therefore, the expression can be rewritten as $\sqrt{x^*Bx + 2\,\mathrm{Re}(y^*Bx) + y^*By}$. Using the fact that $\mathrm{Re}(z) \leq |z|$ and the Cauchy-Schwarz inequality for the B-inner product (which works since conjugate symmetry holds due to hermitian and positive definiteness gives positivity), $|\langle a,\ b\rangle_B| \leq \|a\|_B\|b\|_B = \sqrt{a^*Ba}\sqrt{b^*Bb}$, given that $|\langle a,\ b\rangle_B| = |a^*Bb|$. The expression can further be rewritten as

$$\sqrt{x^*Bx + 2\,\mathrm{Re}(y^*Bx) + y^*By} \leq \sqrt{x^*Bx + 2|y^*Bx| + y^*By} \leq$$
$$\sqrt{x^*Bx + 2\sqrt{y^*By}\sqrt{x^*Bx} + y^*By} = \sqrt{(\sqrt{x^*Bx} + \sqrt{y^*By})^2} = \sqrt{(\|x\|_B + \|y\|_B)^2} = \|x\|_B + \|y\|_B$$

From the expression above, $\|x+y\|_B \leq \|x\|_B + \|y\|_B$ and the triangle inequality hold.

Proof for Non-negativity and definiteness

Since B is hermitian, $(x^*Bx)^* = x^*B^*x^{**} = x^*Bx$, and becuase B is positive definite, x^*Bx is strictly greater than zero and real. for $x \neq 0$. Therefore, $\|x\|_B$ is positive and real for $x \neq 0$ and $\|0\|_B = 0$. Non-negativity and definiteness is upheld.

From the three proofs above, $\|\cdot\|_B$ is a proper norm in $\mathbb{C}^n$.

Problem 2

Given an A-orthonormal system, prove that A is symmetric-positive definite.

Defining a matrix $U \in \mathbb{R}^{n\times n}$ using the column vectors of the A-orthonormal system, $U^TAU = I$, as the multiplication of the A-orthonormal matrices and A should produce the identity matrix. Additionally, If $\sum_i c_i u(i) = 0$, multiplying on the left by $u(j)^TA$ gives $c_j = 0$ for each j. So the columns of U are independent, and U is invertible. By transposing

$$(U^TAU) = I = I^T = (U^TAU)^T = U^TA^TU = U^TAU$$

Because the equation holds, and the fact that U is invertible due to being a square matrix and having linearly independent column vectors, multiplying the prior equation by U^{-T} on the left and U^{-1} on the right gives $A = A^T$. Therefore, A is symmetric. Say for all $x \in \mathbb{R}^n$ and $x \neq 0$, $v = U^{-1}x;\ x = Uv$. $x^TAx = v^TU^TAUv = v^TIv = v^Tv > 0$. Therefore A is positive definite.

Problem 3

Prove that $2Q_A(v) = (e,e)_A - (u,u)_A$, given that $Q_A(v) := \frac{1}{2}(v,v)_A - (b,v)_I$ and $Au = b,\ (b,v)_I = b^T v$

$2Q_A(v) = (v,v)_A - 2(b,v)_I = (v,v)_A - 2(Au,v)_I$. Because $(b,v)_I = b^T v$, $(Au,v)_I = (Au)^T v = u^T A^T v = u^T Av = (u,v)_A$ due to symmetry. Therefore, the expression can be written as $(v,v)_A - 2(u,v)_A$. By completing the square through adding and subtracting $(u,u)_A$, the new expression is

$$(v,v)_A - 2(u,v)_A + (u,u)_A - (u,u)_A = (v-u, v-u)_A - (u,u)_A = (e,e)_A - (u,u)_A$$

.

Problem 4

Part 1

Table 1: Approximate solution vectors obtained with conjugate gradient (CG) and gradient descent (GD) for both matrices $A_1,\ A_2$.

Method	Matrix	Solution vector x^T
CG	A_1	`[0.333, 0.333, 0.333, 0.333, 0.333, 0.333, 0.333, 0.333, 0.333, 0.333, 0.333, 0.333, 0.333, 0.333, 0.333, 0.333]`
CG	A_2	`[0.333, 0.333, 0.333, 0.333, 0.333, 0.333, 0.333, 0.333, 0.333, 0.333, 0.333, 0.333, 0.333, 0.333, 0.333, 0.333, 0.333, 0.333, 0.333, 0.333, 0.333, 0.333, 0.333, 0.333, 0.333, 0.333, 0.333, 0.333, 0.333, 0.333]`
GD	A_1	`[0.333, 0.333, 0.335, 0.333, 0.332, 0.332, 0.332, 0.334, 0.335, 0.335, 0.335, 0.335, 0.334, 0.333, 0.332, 0.330]`
GD	A_2	`[0.333, 0.333, 0.332, 0.334, 0.335, 0.334, 0.334, 0.333, 0.332, 0.332, 0.332, 0.332, 0.332, 0.333, 0.333, 0.334, 0.334, 0.334, 0.335, 0.335, 0.335, 0.335, 0.335, 0.335, 0.335, 0.334, 0.334, 0.333, 0.332, 0.332, 0.331, 0.330]`

Part 2

Table 2: Solutions obtained using different iterative methods.

Method	Approximate solution x^T
Jacobi	`[1.0, -2.0, 3.0, -2.0, 1.0]`
Gauss–Seidel	`[1.0, -2.0, 3.0, -2.0, 1.0]`
Gradient Descent	`[1.0, -2.0, 3.0, -2.0, 1.0]`
Conjugate Gradient	`[1.0, -2.0, 3.0, -2.0, 1.0]`